

Bucket 4 Sizing Method

Inverse-decay pairs · How we decide how big each position should be
Simple guide with a worked example · Production as of July 2026

Bucket 4 shorts inverse ETFs (products that lose value when a stock or sector rises) and hedges them by also shorting the underlying. The goal is to harvest structural decay while keeping crash risk under a hard dollar budget.

One rule to remember: if a name's conditional crash hits, that pair may lose at most 0.75% of the Bucket 4 sleeve budget.

1. The big picture (four steps)

Sizing runs in a fixed order. Later steps can only shrink positions — they never inflate a risky name by redistributing freed dollars.

Step A — Score the book

Rank pairs by expected decay after borrow costs. Convert scores into weights that sum to 100%.

Step B — Cap crash risk

Estimate how much each pair would lose in a crash. Cap gross so that loss $\leq 0.75\%$ of the sleeve.

Step C — Split into two shorts

Turn each pair's gross into an inverse-ETF short and an underlying short using the hedge ratio h .

Step D — Ratchet (inventory guard)

Never blindly cover hard-to-relocate inverse inventory; trim gradually when gross has crept up.

2. Sleeve budget

Bucket 4 gets a fixed share of book target gross (config: `inverse_decay_bucket4.target_weight`, typically ~3%). All pair sizes are fractions of that sleeve budget.

Example sleeve used below: **\$115,500** (illustrative).

3. Step A — Decay / borrow weights

Crash risk is **not** in this score anymore. The score only answers: “How attractive is the decay after expected borrow?”

$$\text{base_score} = \text{decay} / (1 + \text{quad} \cdot \text{borrow}^2) / (1 + \text{linear} \cdot \text{borrow}) / (1 + \text{unc} \cdot \text{borrow_var})$$

Then weights = `base_score / sum(scores)`, clipped to a min/max per name (default 0.5%–35%), then lightly tilted for covariance concentration so highly correlated pairs do not all get large weights at once.

Production knobs (live): `borrow_linear_aversion` = 1.5, `borrow_uncertainty_penalty` = 3.0, `decay_borrow_quad` = 0 (quadratic off because net edge already nets expected borrow).

4. Step B — Conditional crash budget (the risk rule)

For each underlying, as of the run date:

```
runup = max(0, price / 252-day median - 1)
retrace =  $\theta \cdot \text{runup} / (1 + \text{runup})$  with  $\theta = 0.5$ 
tail = worst ~3y 20-day drop + 0.45 · downside vol
C = max(tail, retrace) conditional crash size
```

Intuition: a name that has already run up a lot can still fall a lot from here, even if recent realized crashes look quiet. We take the larger of historical tail and a fraction of the run-up.

Pair loss per \$1 of gross (L):

```
 $L = (1 - h) \cdot \beta / (1 + h \cdot \beta) \cdot C \cdot (1 + \phi \cdot C)$  with  $\phi = 0.5$ 
```

Why this shape: on a crash of size C, the short inverse loses roughly $\beta \cdot C$ on its notional; the short-underlying hedge recovers fraction h of that. With $\text{inv} = \text{gross} / (1 + h \cdot \beta)$, net loss per gross dollar is the formula above. The $(1 + \phi \cdot C)$ term bumps for multi-day compounding on daily-rebalanced inverses.

Cap (trim-only, freed dollars stay in cash):

```
cap_usd =  $p \cdot \text{sleeve\_budget} / \max(L, L\_floor)$ 
gross = min(solved_gross, cap_usd)
 $p = 0.75\%$ ,  $L\_floor = 2\%$ 
```

Important: we do **not** redistribute freed weight into other pairs. Redeploying would concentrate the book and cancel the safety rule.

5. Step C — Split gross into two legs

```
gross = pair_weight × sleeve_budget
inv_short = gross / (1 + h ·  $\beta$ )
und_short = h ·  $\beta$  · inv_short
```

h (hedge ratio) comes from the TR/VCR engine (v7 + v9 tilt): typically between 0.30 and 0.80, centered near 0.45. Higher h = more underlying hedge = less crash beta, but more borrow/margin on the stock leg.

6. Step D — Inverse ratchet (brief)

The inverse-ETF short is grow-only by default: we floor it at $\max(\text{solved}, \text{currently held}, \text{persisted floor})$ so we do not cover inventory that is hard to re-borrow. When held size has crept above the fresh target, a continuous trim gradually closes the gap (paired with the underlying so h stays consistent).

7. Worked example — DAMD / AMD

Numbers below are stylized from a live July 2026 run shape so the arithmetic is easy to follow. Sleeve budget = **\$115,500**. Pair = short DAMD (inverse AMD) + short AMD.

Given inputs

Input	Value	Meaning
Sleeve budget	\$115,500	B4 target gross
Opt2 weight (Step A)	10.9%	Decay/borrow share of sleeve
AMD price vs 252d median	+140% run-up	Stretched vs anchor
Historical tail	51%	Worst 20d drop + downside vol
Hedge ratio h	0.45	Underlying hedge intensity
Inverse beta β	2.0	beta of $-2\times$ inverse
$\rho / \theta / \phi$	0.75% / 0.5 / 0.5	Live crash-budget knobs

Step A — Solved gross (before crash cap)

$$\text{solved_gross} = 10.9\% \times \$115,500 = \mathbf{\$12,590}$$

Step B — Conditional crash C

$$\text{runup} = 1.40$$

$$\text{retrace} = 0.5 \times 1.40 / (1 + 1.40) = 0.5 \times 0.583 = \mathbf{0.292} \text{ (29\%)}$$

$$\text{tail} = 0.51 \text{ (51\%)}$$

$$C = \max(0.51, 0.292) = \mathbf{0.51} \rightarrow \text{historical tail binds}$$

(If AMD had a quiet crash history but the same run-up, retrace would bind instead — that is the “AMD at highs” protection.)

Step B — Pair loss L and the dollar cap

$$\text{unhedged fraction factor} = (1 - 0.45) \cdot 2.0 / (1 + 0.45 \cdot 2.0)$$

$$= 0.55 \cdot 2.0 / 1.90 = 1.10 / 1.90 = \mathbf{0.579}$$

$$L = 0.579 \cdot 0.51 \cdot (1 + 0.5 \cdot 0.51) = 0.579 \cdot 0.51 \cdot 1.255 \approx \mathbf{0.370}$$

Meaning: each \$1 of pair gross loses about **\$0.37** if AMD crashes 51%.

$$\text{Allowed crash loss} = \rho \cdot \text{budget} = 0.0075 \times \$115,500 = \mathbf{\$866}$$

$$\text{cap_usd} = \$866 / 0.370 \approx \mathbf{\$2,340}$$

$$\text{gross_final} = \min(\$12,590, \$2,340) = \mathbf{\$2,340} \rightarrow \text{cut to } \sim 19\% \text{ of the opt2 size}$$

The other $\sim$ \$10,250 stays in cash inside the sleeve. It is not handed to quieter names.

Step C — Two short legs

$$\text{inv_short (DAMD)} = \$2,340 / (1 + 0.45 \cdot 2.0) = \$2,340 / 1.90 \approx \mathbf{\$1,232}$$

$$\text{und_short (AMD)} = 0.45 \cdot 2.0 \cdot \$1,232 \approx \mathbf{\$1,109}$$

Check: $\$1,232 + \$1,109 = \$2,341 \approx$ gross. Check crash loss: $0.370 \times \$2,340 \approx \$866 =$ exactly the ρ budget.

Example summary

Quantity	Before crash cap	After crash cap
Pair gross	\$12,590	\$2,340
DAMD short	~\$6,626	\$1,232
AMD short	~\$5,964	\$1,109
Loss if AMD -51%	~\$4,660	\$866 (0.75% of sleeve)

8. Why this design

- Decay score picks who deserves capital; crash budget decides how much is safe.
- Cap is per-name and dollar-interpretable: “this name may not lose more than p of the sleeve.”
- Trim-only + no redeploy prevents the optimizer from sneaking risk back into the book.
- Hedge ratio h enters L , so a tightly hedged pair is allowed more gross than a loose one with the same crash C .

9. Where it lives in code / config

- Weights: `scripts/v6_b4_pf_weights.py`
- Crash cap: `scripts/b4_crash_budget.py`
- Legs + ratchet: `scripts/bucket4_weekly_opt2.py`
- Hedge/cadence: `scripts/bucket4_hedge_cadence.py`
- Knobs: `config/strategy_config.yml` →
`inverse_decay_bucket4.rules.bucket4_weekly_opt2`
- Telemetry: `data/runs/<date>/b4_crash_budget.csv`

Production reference · ls-algo · Bucket 4 sizing · July 2026